Structural Registry of Nested Geometries

Polygonal, Polyhedral, and Harmonic Registries as Scale-Invariant Incidence Structures

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Working Draft

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Contents

Abstract

This manuscript develops a structural framework in which harmonic intervals arise as measurable consequences of nested geometric incidence rather than as primitive mathematical objects. The construction begins with a common origin and a shared registry surface. Within that registry, regular polygons and regular polyhedra generate incidence structures whose edge lengths, chord lengths, angular separations, diagonals, face relations, vertex relations, and higher-order adjacencies determine families of scale-invariant ratios. These ratios may subsequently admit harmonic interpretation through normalization procedures, but the underlying ratios originate in the geometry of the registry itself.

The central claim does not assert that geometry and harmony are identical. Rather, it establishes a rigorous chain: common registry surface, incidence structure, homogeneous metric quantities, scale-invariant ratios, normalized interval classes, and harmonic interpretation. This order preserves mathematical discipline: construction precedes measurement, measurement precedes ratio, and ratio precedes interpretation. The manuscript therefore provides a compact but rigorous foundation for later work on resonance registries, nested solids, and generalized structural registry theory.

Keywords: structural registry, incidence complex, nested polygons, nested polyhedra, chord ratios, scale invariance, harmonic intervals, ratio registry, constructive geometry.

Notation

Symbol	Meaning
O	distinguished common origin of a registry
$S^1(R)$	circle of radius R centered at O
$S^2(R)$	sphere of radius R centered at O
P_n	regular n-gon embedded in a common circumcircle
V_n	angular vertex set of P_n
θ_n	central angle of a regular n-gon, $2\pi/n$
e_n	edge length of regular n-gon
$c_{n,k}$	k-step chord of regular n-gon
$\rho(x,y)$	ratio x/y
$\mathcal{R}$	incidence complex or registry
V, E, F, C	vertices, edges, faces, and higher-dimensional cells
ι	incidence relation
$\mathcal{M}(\mathcal{R})$	metric registry, the set of homogeneous geometric measurements associated with $\mathcal{R}$
$\mathcal{I}(\mathcal{R})$	ratio registry, the set of dimensionally consistent ratios derived from $\mathcal{M}(\mathcal{R})$
Φ	registry morphism preserving incidence
$T_{P\to Q}$	registry transport from one embedded registry to another
Δ	angular deficit at a vertex of a polyhedron
χ	Euler characteristic, $V - E + F$ for convex polyhedra

1 Motivation and Governing Thesis

Mathematical treatments of harmonic intervals often begin with frequency ratios. Constructive geometry often begins elsewhere: with points, circles, lines, angles, and incidence. This manuscript treats the constructive order as primary and asks how far ratio and harmonic structure can be derived from it.

The governing thesis can be stated as follows:

> A harmonic interval need not begin as a frequency. It can arise as a scale-invariant ratio generated by incidence relations among nested geometric objects.

The explanatory sequence therefore reverses the usual order. Instead of

$$\text{frequency} \longrightarrow \text{interval} \longrightarrow \text{geometry},$$

this framework begins with

$$O \longrightarrow S^1/S^2 \longrightarrow \text{incidence} \longrightarrow \text{measurement} \longrightarrow \text{ratio} \longrightarrow \text{interval} \longrightarrow \text{harmonic interpretation}.$$

The interval becomes an observable projection of prior geometric structure.

2 Foundational Axioms

Axiom 1 (Common Registry Origin)**.** *Every registry possesses a distinguished common origin O serving as the reference from which all registry relations are defined.*

Axiom 2 (Common Registry Surface)**.** *Every object belonging to the same registry shares a common embedding manifold. In two dimensions this manifold is a common circumcircle $S^1(R)$. In three dimensions it is a common circumsphere $S^2(R)$.*

Axiom 3 (Incidence Precedes Measurement)**.** *Adjacency, containment, intersection, and ordering exist before lengths, areas, volumes, or angles receive numerical values. Incidence is therefore primitive; measurement is derived.*

Axiom 4 (Homogeneous Ratio Invariance)**.** *Whenever two geometric measurements possess the same physical dimension, their ratio remains invariant under uniform scaling.*

Axiom 5 (Interpretive Independence)**.** *Registry invariants remain independent of any particular observational interpretation. Musical intervals, frequency ratios, graph spectra, and physical resonances may interpret registry ratios, but they do not create them.*

3 Incidence Complexes and Registry Objects

Definition 3.1 (Incidence Complex)**.** *A registry is represented by an incidence complex*

$$\mathcal{R} = (V, E, F, C, \iota), \tag{1}$$

where V is the vertex set, E the edge set, F the face set, C the set of higher-dimensional cells, and ι the incidence relation.

The planar polygonal registry appears as the special case $C = \varnothing$. The polyhedral registry extends the same incidence structure into three dimensions by allowing C to record enclosed cells. This removes the appearance that polygonal and polyhedral registries belong to unrelated subjects; both become instances of one formal object.

Definition 3.2 (Metric Registry)**.** *The metric registry associated with $\mathcal{R}$ is*

$$\mathcal{M}(\mathcal{R}) = \{m_i\}, \tag{2}$$

where each m_i is a homogeneous geometric measurement associated with the registry. Examples include chord lengths, edge lengths, face diagonals, body diagonals, face areas, surface areas, volumes, dihedral angles, and angular deficits.

Definition 3.3 (Ratio Registry)**.** *The ratio registry associated with $\mathcal{R}$ is*

$$\mathcal{I}(\mathcal{R}) = \left\{ \frac{m_i}{m_j} : m_i, m_j \in \mathcal{M}(\mathcal{R}),\ \dim(m_i) = \dim(m_j),\ m_j \neq 0 \right\}. \tag{3}$$

Only quantities of identical physical dimension are compared. This condition preserves dimensional consistency.

4 Polygonal Registry

Let P_n denote a regular n-gon inscribed in a common circumcircle $S^1(R)$ centered at O. Its central angle is

$$\theta_n = \frac{2\pi}{n}. \tag{4}$$

The angular vertex set is

$$V_n = \left\{ \frac{2\pi j}{n} : 0 \le j < n \right\}. \tag{5}$$

Every vertex, edge, diagonal, and higher-order chord of P_n is determined by this angular registry.

Definition 4.1 (Edge Length of a Regular Polygon)**.** *The edge length e_n of P_n is the chord between adjacent vertices.*

Proposition 4.1 (Polygon Edge Formula)**.** *For a regular n-gon inscribed in $S^1(R)$,*

$$e_n = 2R\sin\left(\frac{\pi}{n}\right). \tag{6}$$

Proof. Adjacent vertices subtend the central angle $2\pi/n$. The two radii to the endpoints of the edge form an isosceles triangle. Bisecting this triangle produces a right triangle with hypotenuse R, angle π/n, and opposite side $e_n/2$. Therefore

$$\sin\left(\frac{\pi}{n}\right) = \frac{e_n/2}{R},$$

which gives the stated formula. □

5 Chord Families

Definition 5.1 (Chord Family)**.** *For $1 \le k \le \lfloor n/2 \rfloor$, the k-step chord $c_{n,k}$ of P_n is the segment joining vertices separated by exactly k registry positions.*

Proposition 5.1 (General Chord Formula)**.** *Every k-step chord length satisfies*

$$c_{n,k} = 2R\sin\left(\frac{k\pi}{n}\right). \tag{7}$$

Proof. The two radii joining the center to the chord endpoints form an isosceles triangle with included angle $2k\pi/n$. Bisecting that triangle produces two congruent right triangles with hypotenuse R, angle $k\pi/n$, and opposite side $c_{n,k}/2$. Hence

$$\sin\left(\frac{k\pi}{n}\right) = \frac{c_{n,k}/2}{R},$$

and the formula follows. □

Corollary 5.1 (Angular Determination)**.** *Every chord family depends exclusively upon registry radius and registry angle. No additional geometric quantity enters.*

6 Ratio Families and Scale Cancellation

Metric quantities become structurally meaningful through comparison. Define

$$\rho(x,y) = \frac{x}{y}. \tag{8}$$

Applying this to two chord families in a common registry gives

$$\rho(c_{n,k}, c_{m,\ell}) = \frac{2R\sin(k\pi/n)}{2R\sin(\ell\pi/m)} \tag{9}$$

$$= \frac{\sin(k\pi/n)}{\sin(\ell\pi/m)}. \tag{10}$$

The registry radius cancels identically.

Theorem 6.1 (Chord Ratio Invariance)**.** *Every ratio formed from chord lengths belonging to a common registry remains invariant under uniform scaling.*

Proof. Under uniform scaling $R \mapsto \lambda R$,

$$c_{n,k} \mapsto 2\lambda R\sin(k\pi/n),$$

and similarly for $c_{m,\ell}$. Therefore

$$\frac{2\lambda R\sin(k\pi/n)}{2\lambda R\sin(\ell\pi/m)} = \frac{\sin(k\pi/n)}{\sin(\ell\pi/m)}.$$

The scale parameter cancels identically. □

7 Nested Angular Registries

Let

$$V_n = \left\{ \frac{2\pi j}{n} : 0 \le j < n \right\}, \qquad V_m = \left\{ \frac{2\pi q}{m} : 0 \le q < m \right\}. \tag{11}$$

The relationship between V_n and V_m is determined by angular coincidence and angular refinement.

Proposition 7.1 (Coincidence Registry)**.** *The shared angular coincidence of V_n and V_m is governed by* $\gcd(n,m)$*.*

Proof. A vertex belongs simultaneously to both registries precisely when

$$\frac{2\pi j}{n} = \frac{2\pi q}{m} \pmod{2\pi}.$$

Equivalently, $jm = qn$ modulo the full rotation. The common periodicity is therefore determined by the greatest common divisor of n and m. □

Proposition 7.2 (Refinement Registry)**.** *The smallest angular registry containing both V_n and V_m possesses* $\operatorname{lcm}(n,m)$ *positions.*

Proof. Every angular subdivision belonging to either polygon appears among the subdivisions of $V_{\mathrm{lcm}(n,m)}$. No smaller regular angular registry contains both sets of subdivisions simultaneously, by the defining minimality of the least common multiple. □

Corollary 7.1 (Polygon Registry Lattice)**.** *The operations* gcd *and* lcm *induce coincidence and refinement relations among polygon registries. Nested polygons therefore organize into a refinement hierarchy rather than an unordered collection.*

8 Polygonal Incidence Graphs

Each polygon determines an incidence graph

$$G(P_n) = (V, E). \tag{12}$$

Vertices correspond to registry positions. Edges correspond to adjacent incidences. Higher-order diagonals define additional edge classes. Consequently, each regular polygon determines not merely one graph but an entire family

$$\{G_1, G_2, \ldots, G_{\lfloor n/2 \rfloor}\}, \tag{13}$$

where G_k joins vertices separated by k registry steps. The complete polygonal registry therefore consists of all admissible incidence graphs generated upon the same angular registry.

9 Polyhedral Registries

The transition from polygonal registries to polyhedral registries extends the common registry from a circumcircle to a circumsphere while preserving the incidence foundation.

Definition 9.1 (Polyhedral Registry)**.** *Let $\mathcal{P} = (V, E, F)$ denote a regular convex polyhedron embedded in $S^2(R)$. The associated registry is*

$$\mathcal{R} = (V, E, F, C, \iota), \tag{14}$$

where C records three-dimensional cells. The polygonal registry is recovered when $C = \varnothing$.

Proposition 9.1 (Polyhedral Metric Determination)**.** *Every regular polyhedron embedded in a common circumsphere determines a family of edge, face, diagonal, angular, and incidence ratios.*

Proof. The common circumsphere fixes the radial distance of every vertex. Once the incidence structure of a regular polyhedron is fixed, all edge lengths, face diagonals, body diagonals, vertex separations, face angles, dihedral angles, and angular deficits are determined up to the common scale R. Ratios of homogeneous quantities therefore depend only on the registry and not on absolute scale. □

10 Platonic Edge Relations

For a shared circumsphere of radius R, the edge lengths of the Platonic solids are:

$$a_T = \frac{4R}{\sqrt{6}}, \tag{15}$$

$$a_C = \frac{2R}{\sqrt{3}}, \tag{16}$$

$$a_O = \sqrt{2}R, \tag{17}$$

$$a_D = \frac{4R}{\sqrt{3}(1+\sqrt{5})}, \tag{18}$$

$$a_I = \frac{4R}{\sqrt{10+2\sqrt{5}}}. \tag{19}$$

Here T, C, O, D, I denote tetrahedron, cube, octahedron, dodecahedron, and icosahedron respectively.

Thus cross-registry edge ratios include

$$\frac{a_T}{a_C} = \sqrt{2}, \tag{20}$$

and

$$\frac{a_O}{a_C} = \sqrt{\frac{3}{2}}. \tag{21}$$

These ratios arise entirely from shared-sphere geometry.

11 Edge-Face-Vertex Relations

For the Platonic solids the triples (V, E, F) are:

Solid	Vertices	Edges	Faces
Tetrahedron	4	6	4
Cube	8	12	6
Octahedron	6	12	8
Dodecahedron	20	30	12
Icosahedron	12	30	20

Each satisfies Euler closure

$$V - E + F = 2. \tag{22}$$

This invariant belongs to the topological registry, not merely the metric registry.

12 Angular Deficit and Vertex Curvature

For a regular n-gon face, the interior angle is

$$\phi_n = \frac{(n-2)\pi}{n}. \tag{23}$$

If q such faces meet at a vertex of a Platonic solid, the angular deficit is

$$\Delta = 2\pi - q\phi_n. \tag{24}$$

For the tetrahedron, $n = 3, q = 3$, so

$$\Delta_T = 2\pi - 3\left(\frac{\pi}{3}\right) = \pi. \tag{25}$$

For the cube, $n = 4, q = 3$, so

$$\Delta_C = 2\pi - 3\left(\frac{\pi}{2}\right) = \frac{\pi}{2}. \tag{26}$$

For the octahedron, $n = 3, q = 4$, so

$$\Delta_O = 2\pi - 4\left(\frac{\pi}{3}\right) = \frac{2\pi}{3}. \tag{27}$$

For the icosahedron, $n = 3, q = 5$, so

$$\Delta_I = 2\pi - 5\left(\frac{\pi}{3}\right) = \frac{\pi}{3}. \tag{28}$$

Curvature itself therefore forms a registry of scale-invariant angular relations.

13 Dual Registries

Duality exchanges vertices and faces while preserving edges:

$$V(P^*) = F(P), \qquad F(P^*) = V(P), \qquad E(P^*) = E(P). \tag{29}$$

Accordingly,

$$\text{Cube}^* = \text{Octahedron}, \qquad \text{Dodecahedron}^* = \text{Icosahedron}, \qquad \text{Tetrahedron}^* = \text{Tetrahedron}. \tag{30}$$

Duality preserves incidence while exchanging complementary registry roles. This demonstrates that registry equivalence need not require metric identity.

14 Incidence Matrices

Metric geometry records lengths. Registry geometry records relationships. Define the vertex-edge incidence matrix

$$B_{VE}(v, e) = \begin{cases} 1, & v \in e, \\ 0, & v \notin e. \end{cases} \tag{31}$$

Similarly define the edge-face incidence matrix

$$B_{EF}(e, f) = \begin{cases} 1, & e \subset f, \\ 0, & e \not\subset f. \end{cases} \tag{32}$$

The combinatorial skeleton is therefore encoded by

$$V \xleftrightarrow{B_{VE}} E \xleftrightarrow{B_{EF}} F. \tag{33}$$

The matrices determine incidence independently of a particular metric realization.

15 Registry Morphisms and Transport

Definition 15.1 (Registry Morphism)**.** *Let*

$$\mathcal{G}_1 = (O_1, S_1, \mathcal{R}_1, \mathcal{M}_1, \mathcal{I}_1), \qquad \mathcal{G}_2 = (O_2, S_2, \mathcal{R}_2, \mathcal{M}_2, \mathcal{I}_2) \tag{34}$$

be two registries. A registry morphism

$$\Phi : \mathcal{G}_1 \to \mathcal{G}_2 \tag{35}$$

is a map preserving incidence relations.

A registry morphism may alter size, orientation, or embedding. Its defining requirement is not metric preservation but incidence preservation.

Definition 15.2 (Registry Transport)**.** *A transport map from registry P to registry Q is a comparison map*

$$T_{P\to Q} : \mathcal{R}(P) \to \mathcal{R}(Q) \tag{36}$$

that associates corresponding structural features and compares their homogeneous measurements.

Examples include

$$T_{P\to Q}(e) = \frac{a(P)}{a(Q)}, \qquad T_{P\to Q}(v) = \frac{|V(P)|}{|V(Q)|}, \qquad T_{P\to Q}(f) = \frac{|F(P)|}{|F(Q)|}. \tag{37}$$

The registry records transformations among shapes rather than only measurements within a single shape.

16 General Scale-Invariance Theorem

Theorem 16.1 (Scale Invariance of Registry Ratios)**.** *Let $\mathcal{R}$ be a registry and let $m_i, m_j \in \mathcal{M}(\mathcal{R})$ possess the same dimensional degree k. Under any uniform scaling $x \mapsto \lambda x$, the ratio m_i/m_j remains invariant.*

Proof. A homogeneous measurement of dimensional degree k transforms as

$$m_i \mapsto \lambda^k m_i, \qquad m_j \mapsto \lambda^k m_j.$$

Therefore

$$\frac{\lambda^k m_i}{\lambda^k m_j} = \frac{m_i}{m_j}. \tag{38}$$

The ratio remains invariant. □

Corollary 16.1. *The theorem applies simultaneously to lengths, areas, volumes, and higher-dimensional homogeneous quantities. The earlier chord-ratio invariance theorem is the length-degree case $k = 1$.*

17 Harmonic Interpretation

A musical-style interval appears when a geometric ratio is normalized into a chosen harmonic window. Given a geometric ratio $\rho > 0$, octave normalization may be written

$$\rho_{\text{oct}} = 2^q \rho, \tag{39}$$

where $q \in \mathbb{Z}$ is chosen so that

$$1 \leq \rho_{\text{oct}} < 2. \tag{40}$$

Geometry supplies ρ. Tuning convention supplies the normalization. Harmonic interpretation therefore follows the registry; it does not define the registry.

18 Structural Registry Theorem

Theorem 18.1 (Structural Registry Theorem)**.** *Let*

$$\mathcal{G} = (O, S, \mathcal{R}, \mathcal{M}(\mathcal{R}), \mathcal{I}(\mathcal{R})) \tag{41}$$

be a registry satisfying the foundational axioms. Every interval derived exclusively from homogeneous geometric quantities within $\mathcal{G}$ is invariant under uniform scaling and depends only upon the incidence structure and metric relations induced by the common registry surface.

Proof. The common registry surface supplies a shared origin and embedding. The incidence complex supplies the structural relations among vertices, edges, faces, and cells. The metric registry assigns homogeneous measurements to these relations. By the scale-invariance theorem, ratios among measurements of equal dimension remain invariant under uniform scaling. Any interval derived by normalizing such a ratio inherits this invariance. Therefore the resulting interval depends upon the registry relation rather than absolute size. □

Corollary 18.1 (Harmonic Projection)**.** *Harmonic interval systems constructed from normalized registry ratios may be interpreted as projections of geometric incidence structures.*

19 Established Results

The manuscript establishes four principal results.

1. A registry consists of a common origin, a common embedding surface, an incidence structure, and associated measurable quantities.

2. Regular polygons and regular polyhedra admit a common incidence-theoretic description through $\mathcal{R} = (V, E, F, C, \iota)$.

3. Homogeneous ratios derived from registry measurements remain invariant under uniform scaling.

4. Harmonic interval systems may therefore be interpreted as normalized ratio systems arising from geometric registries.

20 Conjectural Extensions

The following directions remain outside the established results of this manuscript.

Conjecture 20.1 (Registry Embedding of Interval Systems)**.** *Every normalized interval system generated from homogeneous geometric ratios admits an embedding into a registry generated by nested constructive geometries.*

Conjecture 20.2 (Higher Registry Transport)**.** *Topological structures such as Hopf fibrations may represent higher-order registry transports acting between incidence registries.*

Conjecture 20.3 (General Structural Registry Theory)**.** *Constructive geometry, graph theory, lattice theory, Coxeter systems, sphere packings, and harmonic proportion may admit a common language through registry objects, morphisms, and invariant ratio families.*

These conjectures are retained as future work rather than as results proven here.

21 Conclusion

The central contribution of this manuscript is the construction of a disciplined mathematical pathway from nested geometry to ratio and from ratio to harmonic interpretation. The primitive object is not the frequency, nor the interval, nor the polygon considered in isolation. The primitive object is the registry: a common geometric framework within which incidence relations generate measurable quantities, measurable quantities generate scale-invariant ratios, and ratios admit multiple interpretations.

The governing sequence is

$$\text{Origin} \to \text{Registry Surface} \to \text{Incidence} \to \text{Measurement} \to \text{Ratio} \to \text{Interval} \to \text{Interpretation}. \tag{42}$$

The order matters. Construction precedes observation. Relational structure precedes measurement. Interpretation follows invariant relation rather than defining it.

A Polygon Formulae

For a regular n-gon inscribed in $S^1(R)$:

$$\theta_n = \frac{2\pi}{n}, \tag{43}$$

$$e_n = 2R\sin\left(\frac{\pi}{n}\right), \tag{44}$$

$$c_{n,k} = 2R\sin\left(\frac{k\pi}{n}\right), \tag{45}$$

$$\rho(c_{n,k}, c_{m,\ell}) = \frac{\sin(k\pi/n)}{\sin(\ell\pi/m)}. \tag{46}$$

B Polyhedral Formulae

For a Platonic solid with circumsphere radius R:

$$a_T = \frac{4R}{\sqrt{6}}, \tag{47}$$

$$a_C = \frac{2R}{\sqrt{3}}, \tag{48}$$

$$a_O = \sqrt{2}R, \tag{49}$$

$$a_D = \frac{4R}{\sqrt{3}(1+\sqrt{5})}, \tag{50}$$

$$a_I = \frac{4R}{\sqrt{10+2\sqrt{5}}}. \tag{51}$$

For regular faces:

$$\phi_n = \frac{(n-2)\pi}{n}, \qquad \Delta = 2\pi - q\phi_n. \tag{52}$$

C Registry Summary Diagram